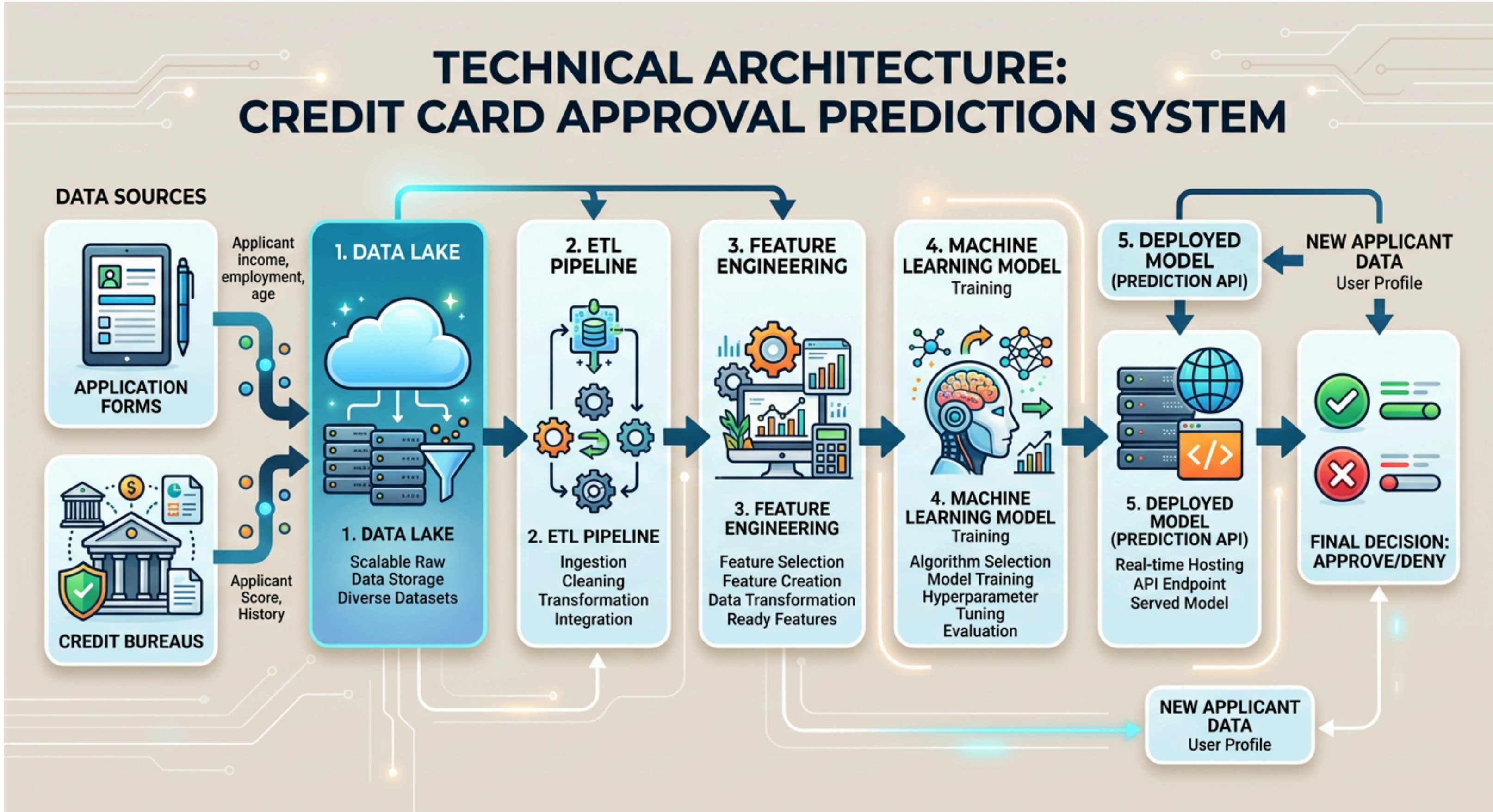
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	30-6-2026
Team ID	23MH1A4257
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Technical Architecture:



TECHNICAL ARCHITECTURE:
CREDIT CARD APPROVAL PREDICTION SYSTEM
DATA SOURCES
APPLICATION FORMS
Applicant income, employment, age
CREDIT BUREAUS
Applicant Score, History
1. DATA LAKE
1. DATA LAKE
Scalable Raw Data Storage
Diverse Datasets
2. ETL PIPELINE
2. ETL PIPELINE
Ingestion
Cleaning
Transformation
Integration
3. FEATURE ENGINEERING
3. FEATURE ENGINEERING
Feature Selection
Feature Creation
Data Transformation
Ready Features
4. MACHINE LEARNING MODEL
Training
4. MACHINE LEARNING MODEL
Training
Algorithm Selection
Model Training
Hyperparameter Tuning
Evaluation
5. DEPLOYED MODEL (PREDICTION API)
5. DEPLOYED MODEL (PREDICTION API)
Real-time Hosting
API Endpoint
Served Model
NEW APPLICANT DATA
User Profile
FINAL DECISION: APPROVE/DENY
NEW APPLICANT DATA
User Profile

S.No	Component	Description	Technology
1.	User Interface	Web application through which users enter loan applicant details and receive loan approval prediction results	HTML, CSS, JavaScript, Bootstrap
2.	Application Logic-1	Handles user inputs, data preprocessing, prediction requests, and displays results	Python, Flask
3.	Application Logic-2	Data preprocessing logic such as handling missing values, encoding categorical features, and scaling data	Python, Pandas, NumPy, Scikit-learn
4.	Application Logic-3	Machine learning inference logic for predicting loan approval status	Scikit-learn (Logistic Regression, Random Forest, Decision Tree)
5.	Database	Stores applicant information, prediction history, and user details	MySQL
6.	Cloud Database	Stores application data in cloud environment for scalability and reliability	IBM Cloudant / IBM Db2
7.	File Storage	Stores trained machine learning models, datasets, logs, and uploaded files	Local File System / IBM Cloud Object Storage
8.	External API-1	Credit score verification or applicant identity	Credit Bureau API (e.g., CIBIL API)

		validation	
9.	External API-2	SMS/Email notification service to notify applicants regarding prediction results	Twilio API / SendGrid API
10.	Machine Learning Model	Predicts whether a loan application should be approved or rejected based on applicant attributes	Logistic Regression, Random Forest, Decision Tree (Scikit-learn)
11.	Infrastructure (Server / Cloud)	Deployment environment for hosting the application	Local Server, IBM Cloud Foundry, Docker, Kubernetes

Table-2: Application Characteristics:

S.No	**Characteristics**	**Description**	**Technology**
1.	Open-Source Frameworks	Frameworks and libraries used for application development and machine learning	Flask, Scikit-learn, Pandas, NumPy, Bootstrap
2.	Security Implementations	Secures application data and prevents unauthorized access	SHA-256 Password Hashing, HTTPS, Flask Session Management, Input Validation, OWASP Security Practices
3.	Scalable Architecture	Three-tier architecture separating presentation, application, and data layers for easier scalability	3-Tier Architecture, Flask REST API
4.	Availability	Ensures uninterrupted application access using cloud deployment and distributed infrastructure	IBM Cloud, Kubernetes, Load Balancer

S.No	Characteristics	Description	Technology
5.	Performance	Optimizes response time through efficient model loading, caching, and lightweight prediction services	Model Serialization using Pickle, In-Memory Caching, Flask WSGI Server

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

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https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d